TÉCNICO LISBOA

# Autonomous Agents and Multiagent Systems

Depending on the courses selected by the students, some other topics can also be considered for the projects, namely: multi-agent reinforcement learning, deep learning, and human-A.I. interactions.

## 2. Organization

### 2.1 Project Proposal and Presentation

A project proposal needs to be submitted (end of week 2), including the plan (limit of 2 pages of text, plus figures and references, using the *ACM Primary Article Template* (2-column article)), and a 15 minute presentation will be done the following week. Sections would normally include;

- An **introduction** that includes motivation, related work, problem definition and relevance, and objectives of the project;
- A description of the **approach** with the specification of the environment, (multi)agent system, and system architecture. This section should also include text to explain why the design choices are adequate to address the problem;
- A description of the **empirical evaluation**. Define a set of metrics that can validate the project's objectives.

For the projects of type "***Reimplement an existing work"*** *this description is simpler has the main focus is on what is going to be replicated.*

### 2.2 Final Delivery

The final delivery (see dates in fenix) with the following elements:

1. **Full source code** and **executable** with a README that explains how to run the system
2. **Final report** (limit of 4 pages of text, plus figures and references, using the *ACM Primary Article Template* (2-column article)).
3. A 1-to-2 minutes **video** demonstrating the agents or algorithms *in-action* (N.B.: a video of text outputs within a terminal (system console) is not accepted).

Be prepared to make a 15-minute presentation of the project (including a presentation of the video) followed by a 5-minute session of questions and answers.

## 3. Project Examples

A computational model for the Downs–Thomson paradox
(https://en.wikipedia.org/wiki/Downs%E2%80%93Thomson_paradox) or related problems

A computational model for the question "Why do competitors open their stores next to one another?" (https://www.youtube.com/watch?v=jILgxeNBK_8&ab_channel=TED-Ed)

A computational model for "The Better Boarding Method Airlines Won't Use"
(https://www.youtube.com/watch?v=oAHbLRjF0vo)

A computational model for "The Simple Solution to Traffic"
(https://www.youtube.com/watch?v=iHzzSao6ypE)

A computational model for all the problems related with elections, e.g. how the minority can win, …
(https://www.youtube.com/playlist?list=PLqs5ohhass_RN57KWlJKLOc5xdD9_ktRg)

4 Tobacco companies a real life example of Game Theory application
(https://www.youtube.com/watch?v=iguBgJP4Cwo&ab_channel=BusinessLearningPresentation)

Game theory challenge: Can you predict human behavior? - Lucas Husted
(https://www.youtube.com/watch?v=MknV3t5QbUc&ab_channel=TED-Ed)

Public Goods Games
(https://www.youtube.com/watch?v=kw9shrf-6U4&ab_channel=SystemsInnovation)

**Papers to reproduce**

Markov games as a framework for multi-agent reinforcement learning
(https://courses.cs.duke.edu/spring07/cps296.3/littman94markov.pdf)

Ad Hoc Autonomous Agent Teams: Collaboration without Pre-Coordination (change to a computational one)
(https://www.cs.utexas.edu/~pstone/Papers/bib2html/b2hd-AAAI10-adhoc.html)

Social diversity and promotion of cooperation in the spatial prisoner's dilemma game
(https://www.semanticscholar.org/reader/f64c6ef4459846b231055fa746b17aa6cab814f5)

Anytime algorithms for multiagent decision making using coordination graphs
(https://ieeexplore.ieee.org/abstract/document/1398426)

Nash Q-Learning for General-Sum Stochastic Games
(https://www.jmlr.org/papers/volume4/hu03a/hu03a.pdf)

Properties of winning Iterated Prisoner's Dilemma strategies
(https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1012644
https://www.youtube.com/watch?v=7wVkXS821ig)

Game Theory, Evolutionary Stable Strategies and the Evolution of Biological Interactions
(https://www.nature.com/scitable/knowledge/library/game-theory-evolutionary-stable-strategies-and-the-25953132/)

Testing Mixed-Strategy Equilibria When Players Are Heterogeneous: The Case of Penalty Kicks in Soccer
(https://pricetheory.uchicago.edu/levitt/Papers/ChiapporiGrosecloseLevitt2002.pdf)